COURSE TITLE : CLIMATOLOGY
COURSE CODE : 3181
COURSE CATEGORY : B
PERIODS/WEEK : 4
PERIODS/SEMESTER : 60
CREDITS : 4

TIME SCHEDULE

Module	Topics	Period
1	Elements and factors of climate	12
2	Comfort factors	18
3	Climate oriented design	18
4	Shelters in tropical climates & Climate responsive design in buildings	12
TOTAL		**60**

Course Objective :

Sl.	Sub	Student will be able to
1	1	understand global climatic factors
	2	Outline climatic elements and characteristics of different types of tropical climates
	3	Understand site climate
2	1	Understand biological approach of climatic design
	2	Know thermal properties of materials
	3	Understand different means of thermal control and their effects in thermal design
3	1	Apply climatic knowledge on design of buildings in different climate
	2	Identify shelters of tropical climates
	3	Apply the principles of solar passive Architecture in buildings.

After completion of course student should be able to:

Module I: ELEMENTS AND FACTORS OF CLIMATE

1.1 .0. Understand global climatic factors

1.1.1 Define climate
1.1.2 Differentiate between weather and climate
1.1.3 Know the quality and quantity of solar radiation
1.1.4 State main cause of seasonal variations
1.1.5 Explain how earth attain thermal balance
1.1.6 Explain global wind pattern
1.1.7 State influence of topography

1.2.0. Outline climatic elements and characteristics of different types of tropical climates

1.2.1 List out various climatic elements
1.2.2 State how these are measured, their units of measurement and instruments used to measure them
1.2.3 Know the data of climatic elements to be collected related to thermal design of buildings
1.2.4 Classify tropical climates
1.2.5 Explain the features of each category of tropical climate

1.3.0. Understand site climate

1.3.1 Differentiate between site climate, macro climate and micro climate
1.3.2 Identify factors which cause deviation from macroclimate
1.3.3 Explain the influence of local factors on temperature, humidity, precipitation, solar radiation and air movement
1.3.4 Explain influence of vegetation on climatic elements

Module II. COMFORT FACTORS

2.1.0. Understand biological approach of climatic design

2.1.1 State the effect of climate on man
2.1.2 Describe the process of thermal balance of body
2.1.3 List out subjective variables which influence individual preferences
2.1.4 Define thermal comfort zone and thermal comfort indices

2.2.0. Know thermal properties of materials

2.2.1 State heat exchange processes
2.2.2 Define Mean radiant temperature, specific heat, latent heat, thermal capacity, U-value, sol-air temperature, and solar heat gain factor

2.2.3 Explain heat exchange processes between buildings and environment
2.2.4 Explain periodic heat flow, time-lag, decrement factor and balancing - in-time.

Module III: CLIMATE ORIENTED DESIGN

3.1.0.Understand different means of thermal control and their effects in thermal design

3.1.1.State objectives of thermal controls
3.1.2.Describe how mechanical controls and structural controls affect thermal design of buildings
3.1.3. Explain the effect of ventilation and air movement on thermal design of buildings

3.2.0.Apply the principles of solar passive Architecture in buildings

3.2.1. Define passive design
3.2.2. Explain passive solar heating strategies like direct gain, thermal storage wall, trombe wall and water wall
3.2.3. Explain passive solar cooling strategies like cool night air,solar chimney, earth coupling, envelope design

Module IV SHELTERS IN TROPICAL CLIMATES & CLIMATE RESPONSIVE DESIGN IN BUILDINGS

4.1.0. Apply climatic knowledge on design of buildings in different climate

4.1.1. Apply the acquired knowledge in thermal design of buildings in hot-dry climate, warm-humid climate, tropical upland climate and composite climate

4.2.0. Identify shelters of tropical climates

4.2.1.Identify the features of form, planning principles , external spaces, planning principles of roofs, walls and openings, surface finishes, ventilation etc..of shelters suitable for hot-dry climate, warm-humid climate, tropical upland climate and composite climate

COURSE CONTENTS

MODULE - I

Climate and tropical climates- Solar radiation – quality and quantity – Tilt of the earth's axis – Radiation at the earth 's surface – Earth's thermal balance –Winds – Trade winds – Mid latitude westerlies – Polar winds – Annual wind shift – Influence of topography - Climatic information – Temperature measurement - Temperature data – Humidity measurement – vapour pressure – Humidity data – Precipitation – Driving rain – Sky conditions – Solar radiation measurement – Solar radiation data – Wind measurement – Wind data – Vegetation - Climatic zones – Tropical climates – Warm humid climate – Warm humid island climate – Hot dry desert climate – Hot dry maritime desert climate – Composite or monsoon climate – Tropical upland climate – Site climate – Deviation within the zone- The designer's task-local factors-air temperature - Temperature inversion – Humidity – Precipitation – Sky conditions – Solar radiation – Air movement – Vegetation – Urban climate -

MODULE – II

Effect of climate on man – body's heat production – heat loss – Regulatory mehanisms – Effects of prolonged exposure – Subjective variables – Definitons of Thermal comfort indices – Effective temperature (E T) – Corrected Effective Temperature (CET) – Bioclimatic chart – Mean radiant temperature – Specific heat – Latent heat - thermal capacity-U-value-sol-air-temperature- solar heat gain factor-heat exchange processes of buildings-thermal balance equation - periodic heat flow-time lag-decrement factor- balancing in time

MODULE-III

Objectives of thermal controls-mechanical controls-degree of controls-heating-problems associated with heating-ventilation-mechanical ventilation systems-cooling by ventilation-evaporative cooling-mechanical cooling-problems associated with cooling-dehumidification-air conditioning-control systems-the need for structural controls-thermal insulation-thermal capacity-solar control-orientation-internal blinds and curtains-heat absorbing glasses-other special glasses-effects of angle of incidence-sun's positions-angle of incidence-shadow angles-shading devices-functions of ventilation –supply of fresh air-convective cooling-provision for ventilation-stack effect-physiological cooling-provision for air movement –wind effects-air flow through buildings-orientation-cross ventilation-size of openings-controls of openings-air movement and rain-air flow around buildings-working of wind-scoop-passive design-direct gain-thermal storage wall-trombe wall-water wall-

MODULE- IV

Shelter for hot-dry climate-nature of the climate-physiological objectives-form and planning-external spaces-roofs , walls and openings-roof and wall surfaces-ventilation and air flow-traditional shelter--shelter for warm humid climates- nature of the climate- physiological objectives-form and planning-external spaces-roofs andwalls-air flow and openings-ventilation- -traditional shelter- -shelter for composite climates- nature of the climate- physiological objectives-design criteria -form and planning-external spaces-roofs andwalls-surface treatment-openings-ventilation and condensation -traditional shelter-shelter for tropical upland climate-nature of climate- physiological objectives-form and planning-external spaces-roofs and walls l surface treatment-openings-traditional shelter

REFERENCES:

1. Koenigsberger, Ingersoll, Mauhew and Szokolay - MANUAL OF TROPICAL CLIMATES
2. R.S. Khurmi & J.K. Gupta - A TEXT BOOK OF REFRIGERATION AND AIR CONDITIONING
3. R.S. Khitoliya - ENVIRONMENTAL POLLUTION – MANAGEMENT AND CONTROL FOR SUSTAINABLE DEVELOPMENT